Sales Performance

Executive Summary

1. Data Quality Issues & Cleaning Process

- **Date Formatting:** The `Invoice Date Key` and `Delivery Date Key` columns were initially recorded as text. They were converted to the standard Date format to ensure accurate Trend Analysis.
- **Missing Values:** 13 rows were found missing the `Delivery Date`. They were kept as is, as they likely represent pending or canceled orders; deleting these rows would result in the loss of their associated revenue data.
- **Unregistered Customers:** It was discovered that 9,077 rows (approx. 34.4% of the dataset) have a value of 0 in the `Customer Key` column. These records were retained as they represent "walk-in" cash sales or unregistered customers. Removing them would cause a massive, unjustified drop in the reported total revenue.

2. Key Business Insights

- **Financial Health:** The company maintains an excellent and stable profit margin of 49.92%, achieving a total profit of $9.92M out of $19.88M in total revenue from 1M units sold.
- **Trend Analysis:** The chart shows a steady upward growth in revenue from 2013, peaking in 2015 (exceeding $6M). The sharp decline in 2016 is not an actual drop in performance, but rather due to incomplete data for that year (the dataset stops at May 2016).

- **Product Performance:** The "Air cushion machine" is the undisputed flagship product, dominating sales and crossing the $1M mark on its own. It is followed by the "Anti static" and "Double sided" product groups.
- **Sales Team Analysis:** Salesperson No. 19 is an exceptional outlier, independently accounting for over 40% of the revenue among the top 5 salespersons (exceeding $1M). This indicates extremely high efficiency or the management of a high-value geographic region/client base.
- **Packaging Preferences:** The vast majority of orders rely on individual packaging (Each) with an overwhelming 87.29% share, reflecting a retail-heavy purchasing pattern rather than wholesale.

3. Recommendations

- **Support Leading Products:** Increase inventory and intensify marketing campaigns for the "Air cushion machine" and its complementary products to ensure a continuous revenue stream.
- **Knowledge Transfer:** Study the sales strategies utilized by Salesperson No. 19 and conduct workshops to transfer these successful tactics to the rest of the sales team, thereby elevating the overall performance average.
- **Supply Chain Optimization:** Since 87% of sales depend on individual packaging (Each), packaging and fulfillment lines should be optimized to be faster and more efficient in handling single-item orders to reduce operational costs.
- **Future Forecasting:** Leverage Machine Learning algorithms to analyze the 2013-2015 data and build a Predictive Model to forecast sales for the second half of 2016. This will assist management in setting accurate targets for the remainder of the year.